MATCS: An Absolute Spatiotemporal Coordinate System for Exact Quantification and Global Localization of Musical Events

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Abstract.

Music has long been regarded as "the art of time" — transient, and hard to measure exactly the way a painting can be. This paper argues that the difficulty does not arise from the essence of music but from its metric: the notational position expression "bar m, beat t" takes as its unit of measurement one beat, which is itself parameterised by the time signature, so that the same expression denotes different physical durations under different signatures. **Change the time signature and the scale changes with it.**

It is necessary first to clarify what this paper does **not** claim. The conversion between bar–beat–tick and absolute tick ($\text{AbsTick} = ((M-1)\cdot \text{BPI} + (T-1))\cdot \text{TPB}$, where $\text{TPB} = \text{PPQ} \times 4/\text{Denom}$) is already standard practice in digital audio workstations, symbolic music libraries and game audio engines (for example the `FBarMap` interface of the Unreal engine, and the time conversions of MusPy and mido), and this paper **does not claim it as new**.

The contribution of this paper is built on that conversion and consists of three items. First, **a unified symbolic-level event criterion**: taking "event = the instant at which the vertical value changes" as the sole definition, so that the detection of note onsets, legato, staccato and rests is reduced to one and the same condition; rests are correspondingly redefined as **empty events** (events whose pitch attribute is empty but whose onset and offset are explicit), so that no undefined interval remains on the time axis. Second, **the closed form under meter change and its degenerate property**: we give a counterexample (the original formula errs by 3840 ticks, exactly one 4/4 bar), generalise the original formula to a piecewise summation form, and prove that the original formula is the degenerate special case of that form when the number of time-signature segments is $n = 1$. Third, **a quantitative characterisation of the applicability domain**: the parameter count of that form grows linearly with the density of meter changes, and as that density approaches "one change per bar" the parameter count approaches the score itself and the formula's leverage drops to zero, at which point the correct answer is not a better formula but a paradigm shift (a time-signature event table + binary search).

On this basis, this paper gives the voice-separation mechanisms derived from these coordinates (coarse register partition, recursive subdivision, the linking method with its "equidistance exclusion method", and a **virtual bridge** that bridges rest gaps), a quantitative form for event relations (legato, staccato and large leaps unified as local quantities that can be counted pair by pair), a precision analysis, and a single observation from a human–AI collaboration.

Scope. This paper does not infer meter from the audio waveform — that assumption cannot be avoided; it does not give a strict upper bound on the rounding error; it does not perform cross-

implementation validation; and it does not evaluate the musicality of the proposed generation architecture.

Keywords: absolute tick; MATCS; spatiotemporal coordinate system; variable time signature; event drawing; empty event; global localization

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1 Introduction

1.1 Background and problem

The processing of symbolic music — editing, analysis, generation and retrieval — all takes "the position of an event on the time axis" as its common reference. In a digital audio workstation (DAW), however, that reference is itself not stable: the position expression used by musical notation, "bar m, beat t", is a **relative** description whose unit of measurement, one beat, is defined by the time signature. Once the time signature changes, the physical duration denoted by the same expression changes with it — **what changes is the ruler, not the thing being measured**.

In engineering practice this property appears as three reproducible classes of failure.

(1) A time-signature switch causes positional drift. The same material edits normally under 4/4, but after switching to 6/8 its notes are systematically displaced and must be corrected by hand one by one. The defect is ubiquitous in existing DAWs, yet the industry has long accepted it as a "normal phenomenon".

(2) Moving material invalidates the coordinates. After material is moved between tracks, the relative relations among its internal notes are unchanged, while its project-level absolute positions all become invalid.

(3) The landing point of generated output cannot be controlled. When a melody produced by a generative model is transferred into MIDI, its landing points are always "close to but not equal to" the target beat positions — by ear one can judge that the intention is bar 2, beat 3, but that intention cannot be reproduced exactly.

The third phenomenon is often attributed to insufficient model capability. That attribution does not hold: generative models perform reliably on formal reasoning of the kind $1 + 1 = 2$, and their failure here shows precisely that **the input representation itself lacks computable precision**, not that their reasoning ability is insufficient.

1.2 Why this is a coordinate problem

Push the analogy towards physical measurement: "the 3rd centimetre of the 2nd meter" is well defined — the ruler is fixed, and anyone measuring at any time obtains the same result. Musical notation is not like that: "bar 2, beat 3" denotes a different tick length under 4/4 and under 6/8.

The substance of the problem is that the unit of measurement of that expression is parameterised by the time signature. When the time signature changes within a piece, the mapping "bar number → tick" is no longer linear but **piecewise linear**. Existing implementations treat the tick as a **counting tool** within one segment, whereas what is needed here is a **coordinate system** that spans segments. The difference between the two is exactly what this paper addresses.

Observation 1.1. *What musical notation lacks is not precision but an absolute ruler that is independent of every variable.*

1.3 Terminological clarification: irregular meter is not variable time signature

Before discussing this problem, a group of terms that have long been conflated must be clarified. Existing accounts often call time signatures such as 5/8 and 7/8 "mixed meter", and conclude on that basis that "complex time signatures are already covered". That classification is misleading in tick mathematics.

Table 1: The properties of four classes of time signature in tick mathematics

Class	Example	BPI	Does it introduce difficulty
Simple meter	2/4, 3/4	2, 3	No
Compound meter	6/8, 9/8, 12/8	6, 9, 12	No (each "beat" is a dotted note)
Irregular meter	5/8, 7/8	5, 7	**No** (unequal division within the bar)
Variable time signature	the signature changes mid-piece	**piecewise values**	**Yes**

The point is this: for the first three classes in Table 1 the BPI is a **constant** throughout the whole piece, and however irregular its value may be (whether $7/8 = 2 + 2 + 3$ or $5/8 = 3 + 2$), the bar length barLen = BPI×TPB never changes, so the coordinate mapping is still linear. What is called "irregular meter" refers to an unequal grouping **inside** the bar; the time signature itself does not change, and the tick mathematics is entirely unaffected by the internal grouping.

Observation 1.2. *What genuinely requires segmentation is* ***variable time signature****, that is, the signature being changed mid-piece. And that is precisely the class that existing validation never covered — the original formula passes on* 4/4, 6/8 *and* 7/4 *and fails only under a changing signature* (*see section 3.7*).

1.4 The contributions of this paper and what it does not claim

It must first be made clear what this paper **does not claim**. The linear conversion between bar–beat–tick and absolute tick is already standard practice in digital audio workstations, symbolic music libraries and game audio engines: the Unreal engine encapsulates it in the `MusicTimestampBarBeatTickToTick` interface of its `FBarMap` class, libraries such as MusPy and mido provide equivalent time conversions, and the commit records of some open-source DAWs even contain entries that specifically fix an "off-by-one" problem in that conversion. **This paper therefore does not claim the conversion as a new mathematical result**; writing it as a closed form merely provides an explicit base for the arguments that follow.

On that base, this paper makes three contributions.

(1) A unified symbolic-level event criterion (section 4). Taking "event = the instant at which the vertical value changes" as the sole definition reduces the detection of the four kinds of boundary — note onset, legato, staccato and rest — to one and the same condition; rests are correspondingly redefined as "empty events", and no undefined interval remains on the time axis. The benefit of this item is testable — four conditions that used to be handled separately are replaced by a single criterion.

(2) The closed form under a changing time signature and its degenerate property (section 3). We give a counterexample (the original formula errs by 3840 ticks, exactly one 4/4 bar), generalise the single linear formula to a piecewise summation form, and prove that the original formula is the degenerate special case of that form when the number of time-signature segments is $n = 1$. The practical significance of the degenerate property is that replacing the implementation changes no behavior of a single-signature project, so it can be deployed safely.

(3) A quantitative characterisation of the applicability domain (section 3). The parameter count of that form grows linearly with the density of time-signature changes; as the density approaches "one change per bar" the parameter count approaches the score itself, and the formula's leverage relative to directly listing the data drops to zero. **Pointing out when a method should be abandoned is as important as giving the method itself.**

On top of this, the paper gives several mechanisms derived from those coordinates — voice separation (section 5.2), the quantification of event relations (section 5.1) and precision analysis (section 3.8). They are **all derived quantities of the coordinate system rather than methods in their own right**; this point is stated again at the beginning of each relevant chapter.

1.5 What this paper does not cover

To avoid misunderstanding, the following are outside the scope of this paper.

1. **It does not infer the beat or the time signature from the audio waveform.** That is a probabilistic estimation problem in Music Information Retrieval; this paper assumes that the segment table is supplied externally, and proposition 2.2 shows that this assumption cannot be avoided.
2. **It does not claim that musical aesthetics can be replaced by numbers.** This paper deals only with position and length, not with "whether it sounds good".
3. **It does not give a strict upper bound on the rounding error under a changing time signature**; only experimental observation is offered (see section 3.9.4).
4. **It does not provide cross-implementation validation.** The experiments were carried out on a single DAW only.
5. **It does not validate the musicality of the proposed generation architecture.** The two worked examples in section 6 verify the consistency of conservation and transformation, not the quality of the resulting piece.

1.6 Organisation of the paper

Section 2 reviews related work and locates the difference between this paper and existing work; section 3 establishes the core coordinate system; sections 4–7 give corollaries at four levels; section 8 discusses the application of these corollaries in human–AI collaboration; section 9 reports the implementation and the experiments; section 10 discusses limitations and boundaries.

2 Related work

2.1 Symbolic music representation

There are two main lines of machine representation for symbolic music. The first is MIDI and its standard file format [1], which records performance information as a stream of event messages and whose time base is the product of "pulses per quarter note" (PPQ) and tempo, which does

[1]MIDI Manufacturers Association, *MIDI 1.0 Detailed Specification*, 1996, [to be verified: edition and cited page numbers]; E. Selfridge-Field (ed.), *Beyond MIDI: The Handbook of Musical Codes*, Cambridge, MA: MIT Press, 1997, p. [X].

not correspond directly to the bar–beat structure of notation. The second is notation description languages such as MusicXML [2] and Humdrum [3], which faithfully describe the visual and structural information of a score, but precisely for that reason their position expressions always remain tied to a time-signature context.

Remark 2.1. Each of the two representations has its strengths, but neither provides an absolute position quantity that is independent of the time signature. What this paper deals with is neither a performance message sequence nor a score document structure, but the **mapping layer** between the two.

2.2 Computational models of beat, time signature and time

The automatic inference of time signatures and beats is a classic problem in Music Information Retrieval (MIR), covering subtasks such as beat tracking, downbeat detection and meter detection, and the relevant methods are mostly based on statistical estimation [4]. Such work differs from the present problem in one fundamental respect: MIR asks "what might the time signature be", which is a **statistical inference** problem; this paper asks "given the time signature, how is a position mapped exactly to a coordinate", which is a **deterministic computation** problem.

This yields a conclusion worth pointing out separately:

Proposition 2.2 (Time-signature information cannot be recovered from ticks)**.** *In raw MIDI or audio data, the time-signature map cannot be recovered uniquely from a tick sequence.*

Proof. The same tick data (for example $0, 480, 960, \ldots, 3360$) is interpreted as 1 bar under 4/4, as 2 bars under 2/4, and as yet other structures under 3/4 and 6/8. Multiple solutions exist, hence there is no unique solution, and the inverse problem is ill-posed. □

Automatic inference of the time signature can therefore give only a statistical estimate and no exact solution, which shows from the other side that the segment table must be **externally supplied prior metadata** and cannot be recovered from the coordinates (see section 3.9 for details).

2.3 Computational musicology toolchains

The field of computational musicology has developed mature toolchains [5]; the formal description of tonal structure is represented by generative grammar theory [6]; and the mechanisms of temporal expectation for musical events have likewise been studied systematically in psychology [7]. Most of this work discusses musical structure within an established metrical context, whereas this paper treats the metrical context itself as the variable to be dissolved.

[2] M. Good et al., "MusicXML: An Internet-Friendly Format for Sheet Music", *XML Conference and Exposition*, 2001, [to be verified: cited page numbers].

[3] D. Huron, "Humdrum and Kern: Selective Feature Encoding", in *Beyond MIDI: The Handbook of Musical Codes*, Cambridge, MA: MIT Press, 1997, pp. [X].

[4] M. Müller, *Fundamentals of Music Processing*, Cham: Springer, 2015, p. [X].

[5] M. S. Cuthbert and C. Ariza, "music21: A Toolkit for Computer-Aided Musicology and Symbolic Music Data", *Proceedings of ISMIR 2010*, 2010, pp. [X].

[6] F. Lerdahl and R. Jackendoff, *A Generative Theory of Tonal Music*, Cambridge, MA: MIT Press, 1983, p. [X].

[7] D. Huron, *Sweet Anticipation: Music and the Psychology of Expectation*, Cambridge, MA: MIT Press, 2006, p. [X].

2.4 Absolute read-outs in DAW software

Modern DAWs generally provide an absolute read-out of the form "bar.beat.tick" (for example REAPER's Big Clock window [8]). That read-out is stable and unique within a project and does not drift with tempo or with other time-signature segments; but when the time signature is changed mid-project, if event coordinates are generated by a single linear formula such as equation (2), a systematic offset appears. The internal implementation of DAW software uses a time-signature event table plus interval lookup, and does not expose a closed-form expression for the case of a changing time signature [9]. One part of this paper's work is to supply that closed form, and at the same time to delimit its applicability boundary.

2.5 Comparison with the standard MIDI tick

The absolute time unit defined in this paper is not the same concept as the tick of the MIDI standard; the differences between the two are given in Table 2.

Table 2: Comparison of the standard MIDI tick with the absolute time unit of this paper

Dimension	**Standard MIDI tick**	**The absolute time unit of this paper**
Relation to tempo	Coupled (a change in BPM directly affects the physical duration)	Decoupled (only the logical density changes)
Relation to the time signature	Implicitly dependent; must be inferred together with tempo and time signature	Explicitly decoupled; computed directly through TPB adaptation
Cross-segment consistency	Inconsistent; segments with different PPQ are not interchangeable	Consistent; forms a unified coordinate system within one project
Mathematical mapping	No direct mapping from "bar.beat" is given	Gives a "minus-one shift + time-signature adaptation" mapping
Systematic architecture	None	Covers single-point localization, global recursion and structural verification

2.6 Limitations of DAW software itself

Worth pointing out separately: the developers of DAW software are clearly aware of the tick inconsistency problem. In documenting its absolute tick input facility, REAPER's official documentation notes that because the software allows MIDI segments with different PPQ to coexist, **a tick in one segment may not be equivalent to a tick in another** [10].

[8] Cockos Incorporated, *REAPER User Guide and ReaScript API Reference*, [to be verified: edition, year and URL].
[9] See the preceding note.
[10] See the preceding note.

This remark delimits exactly the boundary of existing implementations: what they provide is a counting tool **within a segment**, not a coordinate system **unified across segments**. It is the latter that this paper supplies.

2.7 Division of labour with Music Information Retrieval

Beat tracking and onset detection in Music Information Retrieval take the audio waveform as their object and output probability estimates; this paper takes a symbolic event representation (such as MIDI) as its object and outputs deterministic coordinates. The two do not compete: the former answers "what might the time signature be", the latter answers "given the time signature, how is a position mapped exactly". By proposition 1, the former cannot in principle give an exact solution, which shows from the other side that the segment table must be supplied externally.

2.8 Existing bar–beat–tick conversions

This section sets out in one place the part that this paper **does not claim**, so that readers do not misjudge the scope of its contribution. The linear conversion from notational position to absolute tick is already a mature practice in several engineering fields; Table 3 lists some of the implementations for which public documentation exists.

Table 3: Existing bar–beat–tick conversion implementations (not claimed as new by this paper)

Implementation	Form	Notes
Unreal Engine	the `FBarMap` class	Provides the `MusicTimestampBarBeatTick-ToTick` interface: bars and beats are both counted from 1 and ticks from 0, returning a 0-based absolute tick from the start of the piece; the documentation states this convention explicitly
MusPy	symbolic music library	implements the same mapping through a conversion between `resolution` (that is, PPQ) and metrical time
mido	MIDI library	`midifiles/units.py` provides conversion utilities between ticks and time units
madmom	MIR library	its MIDI IO contains an equivalent conversion between ticks and beat positions
the tempo map of an open-source DAW	source commit	commits exist that specifically fix the "off-by-one" problem between bar offset and beat arithmetic

Two points are visible from Table 3. First, the **content** of that conversion is public prior art. Second, even the convention "bars and beats counted from 1, ticks counted from 0" is not this paper's invention — Unreal's interface documentation states it explicitly. This paper therefore claims neither the mathematical content of the conversion nor its counting convention.

What must be distinguished is an **implementation** from a **closed form**. Most of the implementations above work by means of a **time signature map plus interval lookup** (Unreal's documentation states that it is based on a time signature map), and do not expose a closed-form expression for the case of a changing time signature. The work of this paper here is **making explicit** rather than discovery; its value lies not in the formula itself but in the criterion, the degenerate property and the applicability boundary derived from it (section 3).

2.9 How this paper differs from existing work

In summary, this paper does not claim that the quantifiability of music is a new problem; it claims three things. First, the unit "one beat" must be relativised under a changing time signature, and that relativisation has not been written out explicitly in existing accounts. Second, the piecewise summation form that follows from it has a definite degenerate property and can serve as a consistent extension of the original formula. Third, the effective range of that form is constrained by the density of time-signature changes, beyond which a paradigm shift is required. Section 3.9 gives a quantitative characterisation of that boundary.

3 The absolute spatiotemporal coordinate system: a position representation independent of the time signature

This section is the technical foundation of the whole paper. Every conclusion in the sections that follow can be derived directly from the three definitions given here (the TPB conversion base, the absolute tick, and the bidirectional mapping), without introducing any new independent assumption. This section therefore also carries the responsibility of delimiting boundaries: wherever this coordinate system does not hold, the corollaries built on it fail with it, so the applicability domain and the failure mechanism are discussed here as well.

It is worth restating the conclusion of section 2: the conversion in this section is **not a new mathematical result** (see Table 3). The substantive work of this section is twofold — generalising that conversion to a piecewise summation form under a changing time signature and proving that the original formula is its degenerate special case, and giving the quantitative boundary of its applicability domain. Readers interested only in the contributions may go straight to sections 3.7 and 3.9.

3.1 Observation: the stability of the scale

In DAW software, a read-out of the form "bar.beat.tick" remains stable when the time signature, the tempo or the position of the material changes. Table 4 gives the carry rules under three time signatures.

This rule is isomorphic to a positional numeral system: the bar is the high-order digit, the beat the low-order digit, and the numerator of the time signature determines the carry radix. This observation suggests that the position expression can be formalised as a purely counting mapping.

Table 4: Counting carry rules under different time signatures

Time signature	Carry sequence	Carry rule
4/4	1.1.00 → 1.2.00 → 1.3.00 → 1.4.00 → 2.1.00	every four beats carry one bar
6/8	1.1.00 → 1.2.00 → ⋯ → 1.6.00 → 2.1.00	every six beats carry one bar
3/4	1.1.00 → 1.2.00 → 1.3.00 → 2.1.00	every three beats carry one bar

3.2 The conversion base: from PPQ to TPB

Definition 3.1 (Number of numerical coordinate units per beat)**.** Let PPQ be the system's preset time resolution (pulses per quarter note) and Denom the denominator of the current time signature; then the number of numerical coordinate units per beat is defined as

$$\mathrm{TPB} = \frac{\mathrm{PPQ} \times 4}{\mathrm{Denom}}. \tag{1}$$

Equation (1) is the **unit calibration** of every coordinate conversion in this paper: it translates the musicological unit "one beat" into a number of machine-comparable numerical units. Table 5 gives the common values (PPQ = 960).

Table 5: Values of TPB (PPQ = 960)

Time signature	Denom	BPI	TPB	Bar length
4/4	4	4	960	3840
3/4	4	3	960	2880
6/8	8	6	480	2880

Remark 3.2. TPB corresponds one-to-one with the time signature: different signatures correspond to different values, and the same signature corresponds to the same value in any project. It must be stressed that this correspondence presupposes **adopting the DAW's convention for "one beat"** — under 6/8 the DAW defines one beat as an eighth note (TPB = 480), whereas in music theory 6/8 is usually read as two dotted quarter notes. Adopt a different convention for the beat unit and the read-out changes with it (see precondition C in section 3.9).

3.3 The definition of the absolute tick

Definition 3.3 (Absolute tick)**.** Let the start of the project be 1.1.00, let M denote the bar number, T the beat number and BPI the number of beats per bar; then the absolute tick of the position (M, T) is defined as

$$\mathrm{AbsTick}(M, T) = ((M - 1) \cdot \mathrm{BPI} + (T - 1)) \times \mathrm{TPB}. \tag{2}$$

The reasoning behind equation (2) is: $(M - 1) \cdot \mathrm{BPI}$ is the total number of beats before the current bar, $(T-1)$ is the number of beats already elapsed within the current bar, and multiplying by TPB converts the result into a number of absolute ticks.

The two -1 terms in definition 3.3 are not a mathematical convention but the alignment of the counting origins: a notational position is counted from 1, while the tick count starts from

0. Omitting that alignment introduces a systematic bias of "one bar plus one beat", as Table 6 shows.

Table 6: The effect of aligning the counting origins (4/4, BPI = 4, TPB = 960)

Position	**Unaligned**	**Aligned**	
`1.1.00`	$(1 \times 4 + 1) \times 960 = 4800$	$((1-1) \times 4 + (1-1)) \times 960 = 0$	✓
`2.1.00`	$(2 \times 4 + 1) \times 960 = 8640$	$((2-1) \times 4 + (1-1)) \times 960 = 3840$	✓
`1.3.00`	$(1 \times 4 + 3) \times 960 = 6720$	$((1-1) \times 4 + (3-1)) \times 960 = 1920$	✓

Remark 3.4 (The status of the "minus one"). The -1 in equation (2) is a **translation rule** rather than a mathematical rule: it maps a human coordinate counted from 1 onto a machine coordinate counted from 0. Remove that shift and every position moves by "one bar plus one beat" as a whole.

3.4 Event coordinatization

Taking the absolute tick as the horizontal axis of a Cartesian coordinate system and the event pitch P as the vertical axis gives the two-dimensional event coordinate:

$$\text{Event} = (\text{AbsTick},\ P). \tag{3}$$

Every event therefore has a unique start and end absolute coordinate $(\text{Start}_{\text{absTick}},\ \text{End}_{\text{absTick}},\ P)$, whose localization does not depend on the time-signature context. Equation (3) provides a unified mathematical framework for voice separation, melody tracking and harmonic analysis.

3.5 Bidirectional mapping

The inverse mapping can be derived from equation (2). Writing TotalBeats = AbsTick/TPB, we have

$$\text{TotalBeats} = \text{AbsTick}/\text{TPB}, \tag{4}$$

$$M = \lfloor \text{TotalBeats}/\text{BPI} \rfloor + 1, \tag{5}$$

$$T = (\text{TotalBeats} \bmod \text{BPI}) + 1. \tag{6}$$

Equation (2) together with equations (4)–(6) forms a complete inference loop, ensuring that the localization process is unambiguous and traceable.

3.6 Verification

Under 6/8 (BPI = 6, Denom = 8, PPQ = 960, hence TPB = 480) take `4.1.00` for the forward computation:

$$((4-1) \times 6 + (1-1)) \times 480 = 18 \times 480 = 8640.$$

Substituting back into equations (5)–(6) gives a total of $8640/480 = 18$ beats, $M = \lfloor 18/6 \rfloor + 1 = 4$ and $T = (18 \bmod 6) + 1 = 1$, that is `4.1.00`, which agrees with the input. Table 7 summarises the verification data.

Table 7: Bidirectional verification (4/4 and 6/8)

Time signature	Input position	M	T	Absolute tick	Inferred result	Consistent
4/4	`2.3.00`	2	3	5760	`2.3.00`	✓
6/8	`4.1.00`	4	1	8640	`4.1.00`	✓
6/8	`5.3.00`	5	3	12480	`5.3.00`	✓

3.7 A changing time signature: counterexample and generalisation

3.7.1 The implicit assumption of the original formula

Consider the $(M-1)\cdot$ BPI term in equation (2): it uses the **current** BPI **to multiply the total number of bars**. This step assumes that all bars in the project have exactly the same length.

As long as that assumption holds, equation (2) is **exact** (not an approximation), and the verifications above on 4/4, 6/8 and 7/4 are all valid. But as soon as the time signature changes mid-project, the assumption is broken.

3.7.2 Counterexample

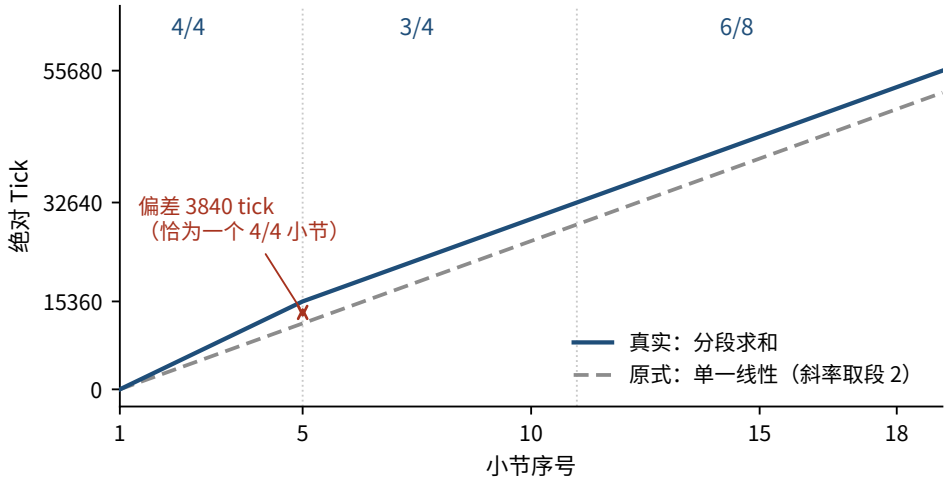


Figure 1: The deviation between the single linear formula and piecewise summation under a changing time signature. The dashed line is the original formula (multiplying by the BPI of segment 2), the solid line is the true piecewise accumulation; the two separate from bar 5 onwards, and the deviation is constantly 3840 ticks, exactly one 4/4 bar.

Consider a project with three time-signature segments, with the parameters given in Table 8.

We want the absolute tick of bar 5, beat 1 (`5.1.00`). Substituting into equation (2) and using the BPI = 3 of segment 2:

$$((5-1)\times 3+(1-1))\times 960 = 4\times 3\times 960 = 11520 \quad \text{(wrong)}.$$

Whereas by piecewise accumulation, the first 4 bars are 4/4 and should give

$$4\times 4\times 960 = 15360 \quad \text{(right)}.$$

Table 8: Settings of the meter-changing project (PPQ = 960)

Segment	Bar range	Time signature	TPB	Bar length
Segment 1	1–4	4/4	960	$4 \times 960 = 3840$
Segment 2	5–10	3/4	960	$3 \times 960 = 2880$
Segment 3	11–18	6/8	480	$6 \times 480 = 2880$

Observation 3.5. *The error is* $15360 - 11520 = 3840$ *ticks,* ***exactly one*** $4/4$ ***bar****. The cause is that the first* 4 *bars were in fact* $4/4$ (4 *beats per bar*) *but were recomputed once by the original formula with the* $\mathrm{BPI} = 3$ *of segment 2.*

The feature that the error is "exactly equal to one bar" deserves attention: it shows that the deviation is not rounding noise but structural — the original formula mistakes a piecewise linear mapping for a straight line.

3.7.3 Mathematical diagnosis

Equation (2) treats "bar number → tick" as a linear mapping solvable by a single multiplication:

$$\mathrm{Tick}(M) = k \cdot M + c \qquad \text{(a straight line)}. \tag{7}$$

But once the time signature changes, that mapping is **piecewise linear** — each segment has a different slope, and the slope of the k-th segment is

$$\mathrm{slope}_k = \mathrm{BPI}_k \times \mathrm{TPB}_k. \tag{8}$$

Observation 3.6 (A key metaphor)**.** *The original formula takes the coordinate function to be a straight line, whereas it is in fact a polyline. For a* ***straight line*** *one multiplication suffices everywhere; for a* ***polyline*** *only* ***piecewise summation*** *works.*

3.7.4 The general form of piecewise summation

Suppose the project is divided into m time-signature segments, with the parameters of the s-th segment as given in Table 9.

Table 9: Notation of the segment table

Symbol	Meaning
n_k	the number of bars governed by segment k
p_k	beats per bar in segment k (= the numerator of the time signature)
d_k	the denominator of the time signature of segment k
t_k	the number of ticks per beat in segment $k = \mathrm{PPQ} \times 4/d_k$
ℓ_k	the number of ticks per bar in segment $k = p_k \times t_k$

The segment start S_s is a cumulative quantity:

$$S_1 = 0, \qquad S_s = \sum_{k=1}^{s-1} n_k \, \ell_k. \tag{9}$$

Writing $A_s = \sum_{k<s} n_k$ for the total number of bars in the preceding segments, the combined general form is

$$\mathrm{AbsTick}(M, T) = \sum_{k<s} n_k \, \ell_k + ((M - A_s - 1)\, p_s + (T - 1))\, t_s. \tag{10}$$

Schematic

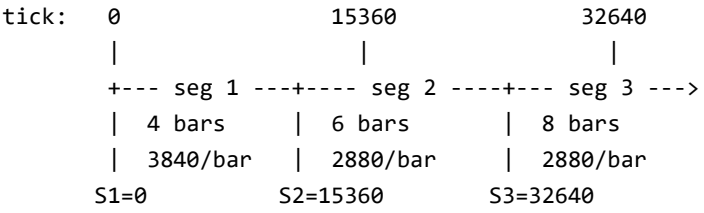


Figure 2: The structure of the time-signature segments. The slope changes at bars 5 and 11, so the cumulative quantity must be summed piecewise.

Remark 3.7 (The demotion of the "minus one"). Compare equation (2) with equation (10): the bar term changes from $(M - 1)$ to $(M - A_s - 1)$. **The "minus one" has not changed, but it has been demoted from a "global minus one" to a "within-segment minus one".**

3.7.5 The degeneracy theorem

Theorem 3.8 (Degeneracy)**.** *When the number of segments is $n = 1$, the general form* (10) *degenerates to the original formula* (2)*.*

Proof. With a single segment $A_1 = 0$ and $\sum_{k<1}(\cdot) = 0$; substituting into (10) gives $\mathrm{AbsTick}(M, T) = ((M - 1) \cdot \mathrm{BPI} + (T - 1)) \cdot \mathrm{TPB}$, which agrees with (2) term by term. □

This conclusion shows that equation (10) is a **domain extension** of the original formula rather than a replacement: the domain of the original formula is $n = 1$ segment, while the domain of the general form is any finite number of segments. In a single-signature project, replacing the implementation changes no existing behavior, so it can be deployed safely.

Remark 3.9 (An assumption becomes a variable). The original formula hard-codes the **assumption** "one and the same time signature throughout the piece" into the formula; the general form turns that assumption itself into a **variable** (the segment table). **Once the assumption becomes a variable, the special case grows into the general form.**

3.7.6 Verification after generalisation

For the project of Table 8, we computed equation (10) point by point and compared the results with measurements from the DAW; the results are given in Table 10.

The hand calculation for `9.2.00` is as follows:

$$\begin{aligned} S_2 &= 15360, \\ \text{within segment} &= (M - A_2 - 1) \cdot p_2 + (T - 1) = (9 - 4 - 1) \times 3 + (2 - 1) = 13, \\ \Delta &= 13 \times 960 = 12480, \\ \text{total} &= 15360 + 12480 = 27840 \ \checkmark. \end{aligned}$$

Table 10: Measured comparison for the meter-changing project

Position	Original formula	General form	Measured in the tool	Verdict
`5.1.00`	11520	**15360**	15360	the general form is correct
`9.2.00`	—	**27840**	27840	the general form is correct
`11.1.00`	—	**32640**	32640	the general form is correct

3.7.7 Rounding error

Equation (1) must be rounded when the denominator is not a power of 2, and the error may accumulate with the number of segments. When $d_k \in \{2, 4, 8, 16, 32\}$ and PPQ = 960 the division is exact and there is no error; for rare denominators rounding is required. It is advisable to choose PPQ as a high power ($960 = 2^6 \times 3 \times 5$, divisible down to /32) in order to reduce the probability of that error occurring. This section gives observation only; a strict proof of the error bound is left to future work (see section 3.9.4).

3.8 Precision analysis

The precision of the absolute coordinate is not determined by human perception but by the numerical resolution of the system. At PPQ = 960, one beat corresponds to 960 ticks, that is 1 tick $\approx 1/960$ beat. Converted into physical time, this gives Table 11.

Table 11: Temporal precision at different resolutions (120 BPM)

PPQ	Time corresponding to 1 tick	Relative to one beat at 120 BPM
960	≈ 0.5208 ms	1/960
9600	≈ 0.052 ms	1/9600
96000	≈ 0.0052 ms	1/96000

Observation 3.10. *At the highest setting this precision already exceeds the resolving limit of human hearing, but its significance lies elsewhere:* ***this precision is determined not by human perception but by the computational precision of the computer.*** *The coordinate system itself has no capability ceiling, only parameter settings.*

For reference, this system precision can be compared with the temporal control ability of human performance. The temporal control precision of a top performer is on the order of $10 \sim 20$ ms; at 120 BPM one beat is 500 ms, so that precision is about 1/50 beat. PPQ = 960 corresponds to 1/960 beat ≈ 0.5 ms, already about 20 times that magnitude; and PPQ = 96000 gives a precision of 0.005 ms, entirely beyond any human perceptual ability.

Remark 3.11. The point of the comparison above is not that "machines are more accurate than people" but rather that, under this system, the source of error is **parameter setting** rather than a **capability ceiling**. Whether higher precision ought to be pursued is a question bound up with aesthetics and use. A note deviating from the standard grid by 0.0.12 can be read either as "not accurate enough" or as "this is simply how he phrases it" — this system is only responsible for measuring it out.

Accordingly, several modifiers in traditional accounts that can be conveyed only by feeling can acquire a quantitative counterpart:

Table 12: From qualitative modifiers to quantitative offsets

Traditional expression	Coordinatized expression
"a little faster"	3 ticks earlier
"a little looser"	2 ticks longer
"not accurate enough"	offset by 6 ticks

Remark 3.12. Table 12 does not mean that aesthetic judgement can be replaced by numbers; it shows only that once position and length are coordinatized, **descriptions of them** can become precise. "Precise" and "pleasant to hear" are two different questions, and this system answers only the former.

3.9 Applicability domain and failure mechanism

This section gives the applicability boundary of the form. Such self-limitation is not common in technical reports, but it is necessary for engineering deployment — a computational interface that does not know when it fails is dangerous.

3.9.1 Time-signature change density

The validity of equation (10) decays with the density of time-signature changes. Table 13 compares three classes of scenario.

Table 13: The applicability domain partitioned by time-signature change density

Scenario	Original formula	General form	Correct solution
A single time signature	correct	correct (degenerate equivalence)	either one
A few meter changes	fails	correct	the general form
Ban-shi change / free meter (sanban) / dense meter change	fails	fails	a time-signature event table + binary search

3.9.2 Parameter complexity

The parameter count grows **linearly** with the time-signature change density; its curve is shown in Figure 3, and Table 14 gives three representative values.

Observation 3.13 (Failure mechanism)**.** *As the time-signature change density approaches "one change per bar", the parameter scale approaches the score itself and the formula's leverage relative to directly listing the data drops to zero.*

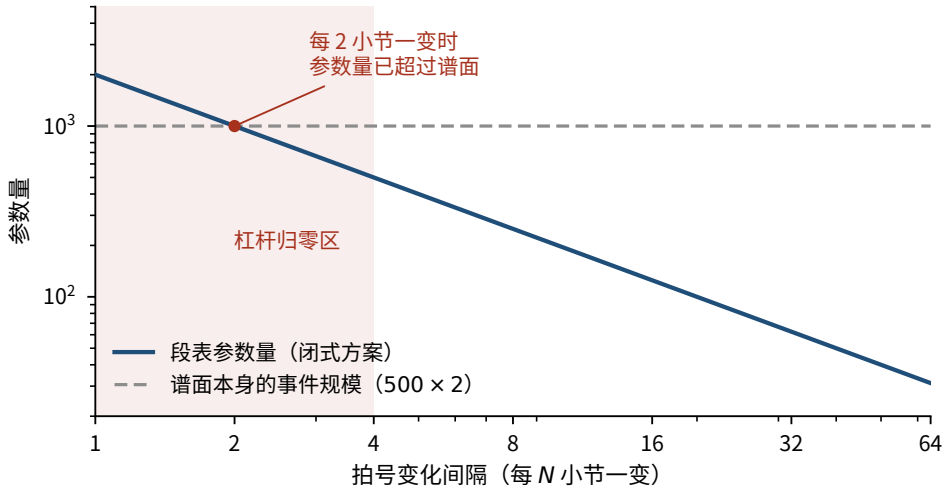


Figure 3: The growth of the parameter count with time-signature change density (a project of 500 bars, on average 2 events per bar). The horizontal axis is "one change every N bars", the vertical axis is logarithmic. The segment-table parameter count is plotted as a continuous value — the actual number of segments must be rounded, and rounding would turn the curve into a staircase, omitted here so that the trend stays visible. As the density rises the parameter count grows with it, and at one change every 2 bars it exceeds the event count of the score itself, whereupon the leverage of the closed form relative to directly listing the data drops to zero.

Table 14: The growth of the parameter scale with time-signature change density

Scenario	**Parameter count**
A single time signature	≈ 3
3 meter-change segments	$3 \times 4 = 12$
dense meter change (one change every 2 bars, 500 bars)	≈ 1000

The correct answer here is not a "better formula" but a **paradigm shift**: store a time-signature event table and change the query to an $O(\log n)$ binary search. This is exactly how mainstream DAWs implement it, and none of them exposes a closed-form formula for the case of a changing time signature [11].

Remark 3.14 (The position of this form). Equation (10) is an $O(1)$ computational interface for the "low-density meter change" scenario. It is not a mathematical discovery, nor a general solution, but a small tool with clearly defined boundaries; its upper bound of value is determined by the time-signature change density.

3.9.3 The irreplaceability of the reverse mapping

The value of equations (2) and (10) is not symmetric between the forward and the reverse direction:

The reason is that the relation between position and tick is a piecewise linear, strictly monotonic bijection, and the reverse solution lands precisely on the cumulative boundaries S_s of the

[11] See the preceding note.

Table 15: Replaceability in the forward and reverse directions

Direction	Form	Can it be replaced by hand
Forward	Position → AbsTick	yes, even by hand calculation
Reverse	AbsTick → Position	**no**, it must rely on the cumulative boundaries S_s

segments; detached from that string of boundaries, the reverse mapping has no unique solution to follow.

3.9.4 Three hard preconditions

The validity of this form depends on the following preconditions; beyond them it does not apply.

1. **Precondition A: the segment table is known.** Equation (10) is not responsible for solving the segment table. By proposition 1, the time-signature map cannot be inferred back from a tick sequence (the problem is ill-posed), so the segment table must be supplied externally (project data or score annotation).
2. **Precondition B: the segment table faithfully reflects the project's time-signature map.** This formula is not responsible for checking it.
3. **Precondition C: the definition of "one beat" agrees with the DAW's convention.** Under 6/8 the DAW may define one beat as an eighth note (TPB = 480), whereas in music theory 6/8 is usually read as two dotted quarter notes. Adopt a different convention and the read-out changes with it.

Remark 3.15. This formula is a calculator, not an auditor: it does not check its preconditions. The accurate statement is this: **given a consistent convention for the beat unit and a correct segment table, the general form holds exactly.**

4 Corollary one: the coordinatization of events

This section sets out the first corollary: once position becomes an absolute coordinate, the question "what is a musical event" is no longer a musicological judgement but reduces to a purely arithmetic condition on the coordinate — **whether the vertical value changes**. Rests consequently no longer need special treatment.

4.1 The event viewpoint and empty events

The notational tradition takes "the presence or absence of pitch" as its basic criterion: sound is a note, silence is a rest. That division introduces unnecessary special handling into the localization problem — a rest becomes a "blank" that needs extra rules to describe.

This paper adopts another viewpoint: music is treated as a time series made up of discrete **events**, each with its own onset and offset times and its own pitch attribute.

Definition 4.1 (Empty event)**.** An **empty event** is an event whose pitch attribute is empty but which has explicit onset and offset times. A rest is one kind of empty event.

Observation 4.2. *If the basic criterion is not "the presence or absence of pitch" but "the onset and offset of the event" as the unified basis for temporal localization, then every musical element (rests included) has a determinate position in the coordinate system.*

Because all events are defined by their onset and offset times, the endpoint of one event coincides with the starting point of the next, forming an uninterrupted **event chain** in the continuous-time representation. The junction between any two adjacent events constitutes the unique event boundary at that instant.

Observation 4.3. *From the event viewpoint there is no "blank" on the musical time axis — every instant belongs to some event, and no ambiguity or undefined region exists.*

4.2 The event criterion

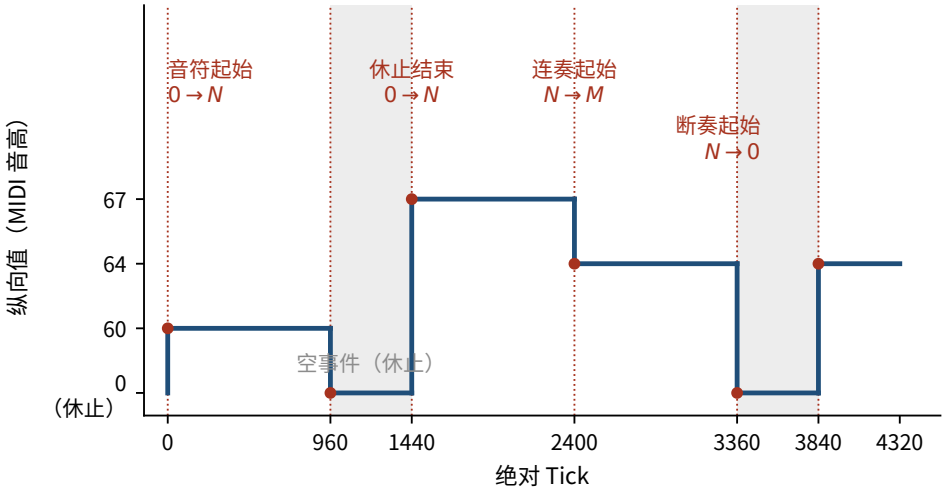


Figure 4: The geometric meaning of the event criterion. The vertical axis carries the equally spaced rows of vertical values (labeled with the true MIDI pitches), the horizontal axis is the absolute tick. Every step of the staircase is one event; the four kinds of boundary (note onset, legato onset, staccato onset, rest end) are triggered by the same condition. The grey interval is an "empty event", that is, a rest.

Once continuous time is discretised by absolute ticks, there is a unique vertical state on each time slice. The event criterion follows from this:

Definition 4.4 (Event)**. An event = the instant at which the vertical value changes.** The vertical value here is the absolute MIDI pitch, and a value of 0 denotes a rest.

This criterion reduces four phenomena that are distinct in notation to one and the same detection logic, as Table 16 shows.

Table 16: The four classes of event expressed under one and the same criterion

Event type	**Change in the vertical value**	**Additional condition**
Note onset	$0 \to N$	—
Legato onset	$N \to M$	N has not yet ended at time T (there is an overlap)
Staccato / rest onset	$N \to 0$	—
Rest end	$0 \to N$	belongs to the same class as the note onset

4.3 Event-chain recursion

The junction property of the event chain yields the recursion relations:

$$E_1^{start} = \mathrm{AbsTick}(M_1, T_1), \tag{11}$$

$$E_i^{end} = E_i^{start} + \delta_i, \tag{12}$$

$$E_{i+1}^{start} = E_i^{end}, \tag{13}$$

where δ_i is the duration of the i-th event. Equation (13), "the end of the previous event equals the start of the next", is not a proposition of music theory but a direct corollary of the continuity of the time axis under discrete coordinates.

Single-point localization is knowing the position of one particular note; **global localization** is starting from the first note and generating all events in one pass by means of the recursion (13). The difference between the two is only whether that recursion is applied.

4.4 The full-scan algorithm

In implementation the logic above is equivalent to a single full scan from the start of the time axis to its end:

```
def scan_events(states):
    """states[t] = the vertical value at the t-th absolute tick (0 means a rest)"""
    events = []
    for t in range(1, len(states)):
        if states[t] != states[t - 1]:          # the vertical value changes
            events.append((t, states[t - 1], states[t]))
    return events

def classify(prev, cur):
    if prev == 0 and cur != 0: return "onset"
    if prev != 0 and cur == 0: return "rest"
    if prev != 0 and cur != 0: return "legato"
```

5 Corollary two: the coordinatization of articulation and voices

This section sets out the second corollary: legato, staccato, large leaps and voice membership all need no separate definition — they are direct read-outs of coordinate differences. Writing

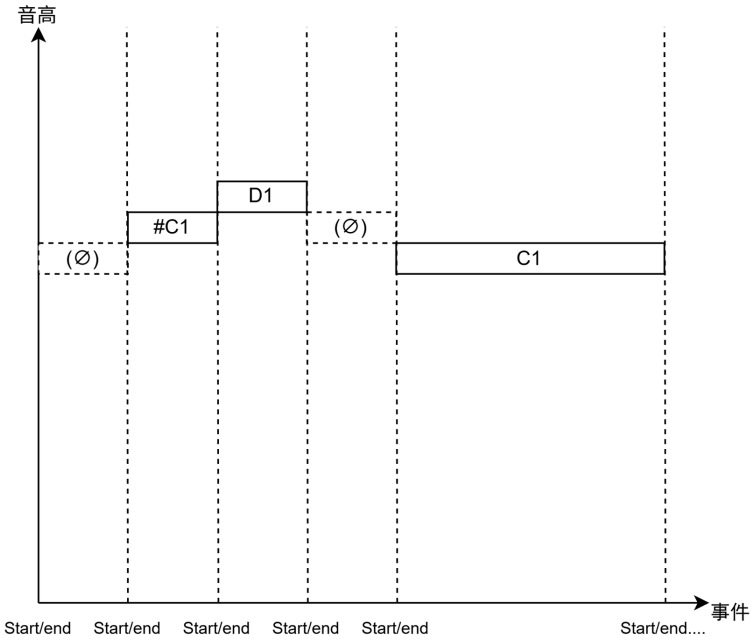
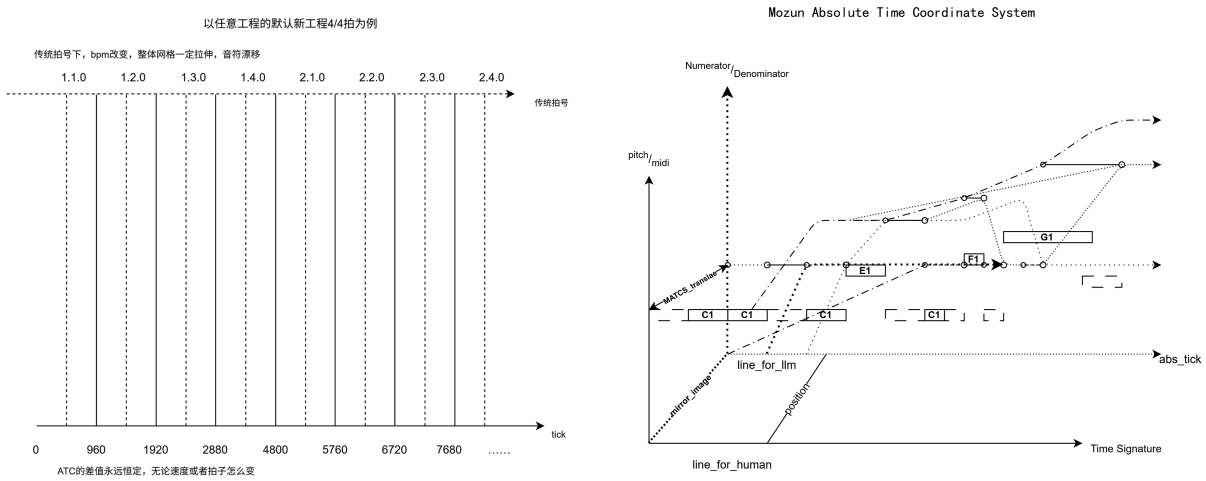


Figure 5: Event drawing. The solid blocks are note events, the dashed blocks are empty events (rests), and the two share one and the same onset/offset structure.



(a) the absolute tick scale

(b) vertical mapping and read-out

Figure 6: The spatiotemporal coordinatization of musical events. The horizontal axis supplies the unique scale, the vertical axis supplies the event criterion.

them as independent methods is misleading; the accurate statement is that **they are all derived quantities of the core coordinate system**.

5.1 The quantification of event relations

5.1.1 A unified criterion

With absolute coordinates available, qualitative descriptions such as "legato" and "staccato" can acquire a unified quantitative form. Let A and B be adjacent events and define the difference

$$\Delta = A_{\text{tick}}^{end} - B_{\text{tick}}^{start}. \tag{14}$$

Then

$$\Delta > 0 \;\Rightarrow\; \text{legato (overlap)}, \qquad \Delta = 0 \;\Rightarrow\; \text{perfect connection}, \qquad \Delta < 0 \;\Rightarrow\; \text{staccato (gap)}. \tag{15}$$

The magnitude $|\Delta|$ allows a further grading, as Table 17 shows.

Table 17: The grading of legato and staccato (PPQ = 960)

Degree	**Range of $\lvert\Delta\rvert$**	**Notes**
Slight	$0 \sim 120$ ticks	within about one eighth of a beat
Marked	$120 \sim 480$ ticks	within half a beat
Strong	> 480 ticks	more than half a beat; under legato it forms a suspension

5.1.2 Large leaps

Let the pitches of two adjacent events be Y_1 and Y_2; then the criterion for a large leap is

$$|Y_2 - Y_1| \geq 5 \;\Rightarrow\; \text{large leap}. \tag{16}$$

The threshold 5 corresponds to a perfect fourth, that is, intervals within a third (≤ 4 semitones) do not count as large leaps. Table 18 gives the verdicts for common intervals.

Table 18: Intervals and the large-leap verdict

Interval	**Number of semitones**	**Verdict**
within a third	≤ 4	not a large leap
perfect fourth	5	large leap
augmented fourth / diminished fifth	6	large leap
perfect fifth	7	large leap
minor sixth	8	large leap
an octave or more	≥ 12	large leap

5.1.3 Two-dimensional coupling

Equations (15) and (16) characterise the horizontal and the vertical dimension respectively; together they classify adjacent events into four classes, see Table 19.

Table 19: Joint classification by the horizontal difference and the vertical pitch difference

Horizontal (Δ)	**Vertical ($\lvert Y_2 - Y_1 \rvert$)**	**Verdict**
overlap > 0	≥ 5	legato large leap
overlap > 0	< 5	legato stepwise motion
gap > 0	≥ 5	staccato large leap
gap > 0	< 5	staccato stepwise motion

Remark 5.1. The significance of this classification is that it decomposes an overall impression such as "how does this melody rise and fall" into local quantities that can be counted event pair by event pair. The case study in section 8 uses exactly this decomposition to compare the differences between two melodies.

5.2 Voice separation and melody line tracking

5.2.1 Defining the problem

Voice separation is a prerequisite shared by melody tracking, harmonic analysis and structured editing. The traditional approach relies on a register prior (for example "the right hand is higher than the left"), but that prior fails where voices cross.

Definition 5.2 (The relativity of voice distinction)**.** The distinction between voices is not absolute (for example "the right hand of a piano is always above C4") but relative — it depends on the specific pitch range of the current passage of notes.

This definition means that voice separation ought to be an **adaptive, unsupervised partition** problem rather than a fixed-threshold classification problem. This paper gives three progressively refined criteria: coarse register partition (5.2.2), recursive subdivision (5.2.3) and temporal continuity (5.2.4).

5.2.2 Coarse register partition

On the vertical axis a four-part choir (SATB) usually has a clear stratification of registers. Let Y' be the vertical coordinate after octave normalisation; then a single simple threshold suffices for the coarse partition: Y' above the threshold goes to the upper voice, and those below the threshold to the lower voice.

Take a passage of piano music as an example: the right-hand melody lies in the middle-high register and the left-hand accompaniment in the middle-low register. With $Y' = -0.5$ as the threshold, the right-hand notes ($Y' \in \{0, +1\}$) and the left-hand notes ($Y' \in \{-1, -2\}$) can be sorted into two voice lists.

Remark 5.3. In essence this coarse partition slices a two-dimensional point lattice along the Y' axis by "contour lines". It fails when octave crossing occurs in four-part harmony, and must therefore be supplemented by the criteria that follow.

5.2.3 Recursive subdivision

To escape the fixed threshold, a recursive bisection strategy can be used: first locate the globally highest and lowest notes, take their midpoint and divide the pitch space into an upper and a

lower half; then take the midpoint of each half in turn, ending with three intervals. The process is equivalent to an adaptive, unsupervised clustering.

Table 20: The three progressively refined voice criteria

Level	Basis	Applicable	Failure condition
Coarse register partition	a threshold on the Y' axis	clear separation of registers	octave crossing between voices
Recursive subdivision	valleys in the pitch-distribution density	no register prior	severe overlap of two registers
Temporal continuity	the nearest distance from a note offset to a note onset	voice crossing	rest gaps must first be resolved

5.2.4 The linking method

Definition 5.4 (The linking method)**.** At the offset of any note, that note's offset is automatically matched with the onset of the currently **nearest** note as one voice, and a line is drawn. If the two overlap there is nothing to consider: they are necessarily the same voice. If one note is at the same distance from both the upper and the lower voice of the preceding note, then by reversing this logic the voice in question must be the middle voice.

The criterion consists of two rules.

Rule one: match the note offset with the nearest onset. For a note A, scan all note onset positions after it and take the nearest as its successor: if there is an overlap (A's offset is later than the successor's onset), they are necessarily in the same voice; if there is no overlap but the distance is the smallest, they are likewise assigned to the same voice.

Remark 5.5. In polyphonic music the notes of one voice usually meet end to end or overlap slightly, and very large time gaps are rare. Judging by "nearest distance" is far more accurate than judging by "whether they lie in the same register" — because when voices cross the upper voice may temporarily lie below the middle voice, but **the temporal continuity is not broken**.

Rule two: the equidistance exclusion method. If the distance from the onset of note C to the offset of the upper note A equals its distance to the offset of the lower note B, then C belongs neither to the upper voice nor to the lower voice and must be the middle voice.

Observation 5.6. *The "equidistance exclusion method" resolves the grey zone of voice membership: when a note sits exactly between the upper and the lower voice, it is not judged by pitch but inferred by reversing the equality relation of the distances. Its mathematical meaning is that if C is equidistant from A and B, then C is independent of the linking relations of A and B on the time axis and must belong to a third, independent voice.*

The workflow in which the two rules cooperate is: traverse from the first note; for each note's offset find the nearest onset; if they overlap, lock the same voice directly; if equidistant, infer the middle voice; once the lines are connected, output the melodic line.

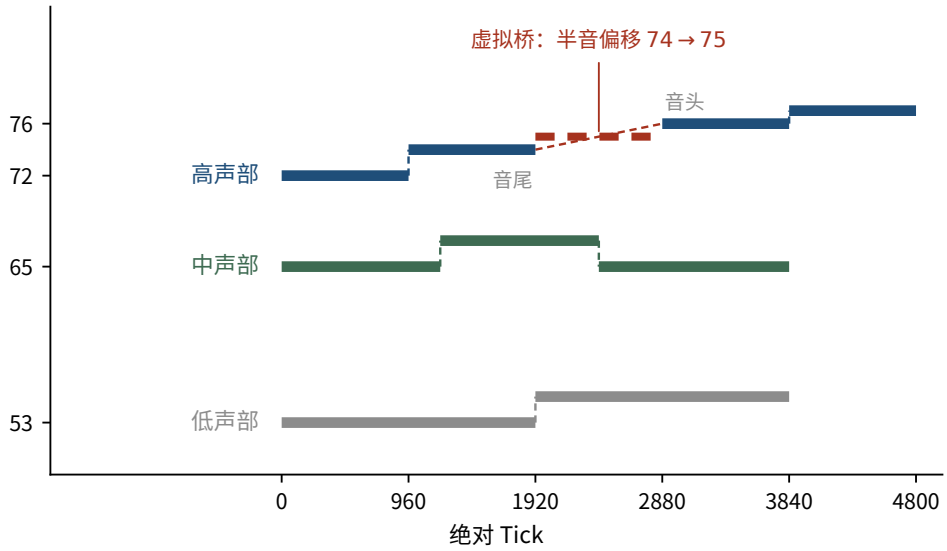


Figure 7: Voice separation and the virtual bridge. The three voices are shown as horizontal bars, the dashed lines are the "note offset to nearest onset" links. The upper voice has a rest gap between ticks 1920–2880, so the link is interrupted; after a virtual note offset by a semitone in pitch bridges it (the red dashed bar), the link runs through; when the tracking is finished the bridge is dismantled, and the interval it occupied is recorded as a rest.

5.2.5 The virtual bridge: bridging rest gaps

The linking method relies on temporal continuity, and a rest leaves a gap in the event chain that interrupts the tracking. The treatment adopted in this paper is to introduce a **virtual bridge**.

Definition 5.7 (Virtual bridge)**.** At a rest, assume a virtual note that starts at the current note offset, takes its value at a **semitone offset** and continues to extend until it meets the next real note and then disappears, so that the linking method can naturally complete the connection "real note → virtual note → real note".

A semitone offset is used rather than the same pitch because the same pitch cannot overlap (a limitation of the MIDI protocol and of physical performance). If the virtual note had the same pitch as the real note, the linking method would run into a logical conflict; a semitone offset (such as C2 → #C2) both avoids the same pitch and preserves the smallest possible interval difference, so it does not affect the judgement of voice membership.

Observation 5.8 (Dismantling the bridge after crossing)**.** *The sole task of the virtual note is to "fill" that stretch of silence with its length, so as to connect the gap in the algorithm. Once the tracking is complete, the algorithm determines that "this is a virtual note and is not output to the final result". But* ***the exact span of time it originally occupied is preserved untouched*** *— that span is precisely the natural silence between the end of one note and the start of the next.*

A rest therefore needs no additional special handling: it is bridged, tracked and then dismantled, and the span of length left after dismantling constitutes the rest event. This is consistent with the position taken in section 4, where a rest is defined as an empty event — a rest is not "nothing" but a stretch of time with an onset and an offset.

6 Corollary three: the coordinatization of structure

This section sets out the third corollary: the sectional structure of a piece can be represented as a recursive division of intervals on one and the same coordinate axis, and motivic development as a geometric transformation on that coordinate plane. Both are coordinate operations and require no "understanding" of the music whatsoever. Absolute coordinates answer the question "where is the event". A natural question follows: can the same set of coordinates be used in the reverse direction, for generation, that is, "where should the event be". This section gives a three-layer architecture whose design principle is to handle **what can be generated** and **what is not allowed** separately.

6.1 The three axioms

1. **Total temporal conservation.** Let the total duration of the piece be T_{total} (measured in ticks), divided into n sections, with the i-th section of duration S_i; then
$$T_{\text{total}} = \sum_{i=1}^{n} S_i. \tag{17}$$
2. **Recursive division.** Any section can be divided further: $S_i = \sum_j s_{ij}$.
3. **Structural template.** The organisation of the sections of a piece is a sequence pattern: Structure = pattern$(S_1, \dots, S_n)$.

Remark 6.1. Axiom (17) and the coordinate definition of section 3 are two ways of saying the same thing: the former constrains the total length, the latter gives the position of each event. Together they make "dividing the total time arbitrarily" a computable operation.

6.2 The structural layer: recursive division under temporal conservation

The legitimacy of a structure is guaranteed by equation (17). Taking common musical forms as examples, the concrete form of the constraint is given in Table 21.

Table 21: Temporal conservation constraints for common musical forms

Musical form	Constraint
A–B–A	$T = S_A + S_B + S_A$
A–B–C–A	$T = S_A + S_B + S_C + S_A$
A–B–A–B–A	$T = S_A + S_B + S_A + S_B + S_A$

This constraint accommodates both square and non-square structures: $S_A = S_B = S_A = 16$ and $S_A = 13,\ S_B = 17,\ S_A = 18$ give the same $T = 48$. In other words, conservation constrains only the total, not the distribution; whether a distribution is reasonable is an aesthetic question rather than a mathematical one.

Theorem 6.2 (Recursion of musical form)**.** *Any musical-form structure can be recursively subdivided into smaller musical-form structures, and the recursion can proceed indefinitely.*

Proof. By induction on the number of layers k. Base case $k = 1$: a single section satisfies $S_A = \sum_j s_{aj}$ by itself, that is, one section can be divided into several subsections. Inductive step: assume the k-th layer holds, that is $p_i = \sum_j p_{ij}$; then repeating the same argument for each p_{ij} gives layer $k + 1$. Hence one can recurse from the division of a movement down to periods, phrases and motives, and finally to individual notes. □

This theorem means that the *A–B–A* structure of a symphony and the *a–b–a′* structure of a 16-bar melody are mathematically isomorphic.

6.3 The generative layer: geometric transformations of the motif

Definition 6.3 (Motif)**.** A motif is an ordered sequence of notes with coordinates $M = \{n_1, \dots, n_k\}$, where each note $n_i = (x_i, y_i)$, x_i is its starting absolute tick and y_i its MIDI pitch number.

Under the two-dimensional coordinates of equation (3), all the traditional techniques of motivic development can be written as unified geometric transformations, see Table 22.

Table 22: The geometric transformation form of motivic development techniques

Technique	Transformation	Dimension affected
Repetition	$n_i' = (x_i + \Delta x,\ y_i)$	temporal translation
Inversion	$n_i' = (x_i,\ 2Y_{\text{ref}} - y_i)$	vertical reflection about the axis Y_{ref}
Retrograde	$M' = \{n_k, n_{k-1}, \dots, n_1\}$	sequence reversal
Augmentation	$n_i' = (x_i \cdot \lambda,\ y_i),\ \lambda > 1$	temporal scaling
Diminution	$n_i' = (x_i \cdot \lambda,\ y_i),\ 0 < \lambda < 1$	temporal scaling

Transformations can be composed; for example $M' = \text{scale}_x(\text{reflect}(M, Y_{\text{ref}}), 2.0)$ means invert first and then augment twofold.

Observation 6.4. *Under the coordinate representation, motivic development requires no "understanding of music", only the execution of algebraic operations. There is no essential difference from other reasoning tasks.*

6.4 The filtering layer: making constraints computable

The generative layer is responsible only for "this arrangement is possible", not for "that arrangement is not allowed". The latter is the task of the filtering layer.

Theorem 6.5 (Existence of a filtering function)**.** *Given any musical constraint C, there exists a computable filtering function F_C such that $F_C(\text{sequence})$ returns true if and only if that sequence satisfies C.*

Proof sketch. The constraint C need only be decidable to be written as a predicate on sequences; checking the elements of the sequence one by one yields F_C. This construction guarantees existence, though not efficiency. □

Table 23 lists several common constraints and the form of their filtering functions.

Table 23: Common constraints and their corresponding filtering functions

Constraint	**Filtering function**	**Effect**
Tonality	$p \bmod 12 \in C_{\text{major}}$	ensures that notes fall within the specified scale
Range	$y_{\min} \le y \le y_{\max}$	ensures that the instrument's range is not exceeded
Legato	$\neg(\Delta > 0 \wedge y_i = y_{i+1})$	excludes overlaps of the same pitch
Harmony	$p \in$ chord template	constrains vertical simultaneity
Proportion	$S_i/S_j \in [r_{\min}, r_{\max}]$	constrains the proportions of sections

Remark 6.6. The filtering condition "the same pitch cannot overlap" has the same reason as the semitone offset used by the virtual bridge in section 5.2.5: it is at once a physical constraint and a constraint of algorithmic consistency.

Multiple constraints are combined by conjunction:

$$F_{\text{total}} = F_1 \wedge F_2 \wedge \cdots \wedge F_k. \tag{18}$$

Any sequence that passes all the filters of equation (18) satisfies all the constraints by construction. A new requirement needs only one additional filtering function, with no change to the generative layer.

6.5 The closed loop and the overall form

The relation among the three layers is: the structural layer decides "how long and into how many sections", the generative layer decides "how the notes are arranged", and the filtering layer decides "which notes cannot be used". The three form a closed loop — if a generated result is rejected by the filtering layer, it falls back to the generative layer for a new transformation, while the total-duration constraint always remains unchanged. On this basis, music generation can be written as

$$\begin{aligned} \text{generation} = {} & \underbrace{\text{recursive division under temporal conservation}}_{\text{structural layer}} \\ & + \underbrace{\text{geometric transformation}}_{\text{generative layer}} \\ & + \underbrace{\text{filtering-function constraints}}_{\text{filtering layer}}. \end{aligned} \tag{19}$$

6.6 Reverse verification: structural boundary alignment

The coordinate definition of section 3 is the forward mapping "from position to coordinate". This section gives a reverse check: using macro-structural constraints to verify the consistency of the coordinate system.

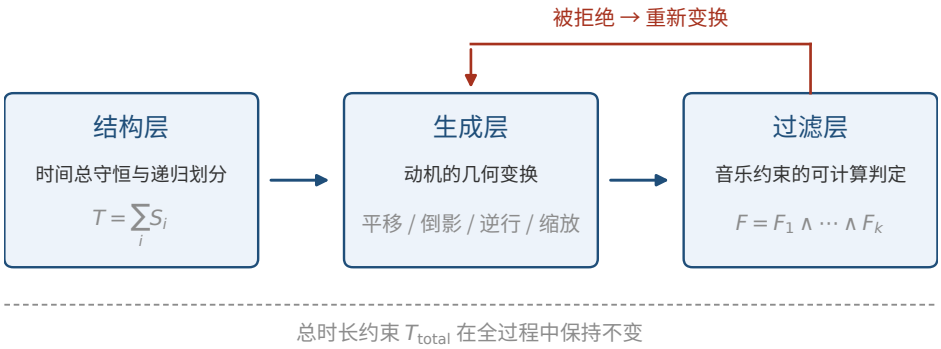


Figure 8: The structure–generation–filtering three-layer closed loop. The structural layer constrains the division into sections by temporal conservation, the generative layer produces motivic variants by geometric transformation, and the filtering layer decides whether a result satisfies the musical constraints; a rejected result falls back to the generative layer for a new transformation, while the total-duration constraint remains unchanged throughout.

The basic idea is to use the macro musical-form structure of a piece to divide the time axis recursively into subspaces, and to use that as a higher-order verification environment in order to prove in reverse that the coordinate system is exact under complex structures. The argument has the structure of a reductio:

1. **Assumption.** Suppose the coordinate formula has an error, or is coupled to the time signature or the tempo in a way that has not been noticed;
2. **Consequence.** Then when the macro boundaries are divided recursively, the accumulated effect of the error means that the boundaries between the sections A, B and A' cannot possibly align exactly and will show up as overlaps or gaps;
3. **Test.** "Whether the boundaries align exactly" is a decidable proposition;
4. **Conclusion.** If the boundaries at every level all align exactly, the coordinate system holds at the macro level.

Formally, let the total duration be T_{total} and the set of sections be $\{S_i\}$; the verification then asserts

$$\sum_i S_i = T_{\text{total}}, \tag{20}$$

and applies the same assertion recursively to every S_i.

Remark 6.7. This verification is effective because it places the test points on the **boundaries** rather than at arbitrary points: the error of a whole segment accumulates at a boundary, which makes the test far more sensitive than a single-point check. This also explains the phenomenon in section 3.7 where the error is "exactly equal to one bar" — that is precisely the typical manifestation of a boundary misalignment rather than random rounding.

6.7 Two worked verification examples

Worked example one: 48 bars, *A*–*B*–*A*. The parameters are 4/4, PPQ = 960 and a total length of 48 bars, hence $T_{\text{total}} = 48 \times 3840 = 184320$ ticks. The seed motif is $M = \{\text{C4, D4, E4, F4, G4}\}$. Section B uses inversion, with the axis of reflection at $Y_{\text{ref}} = 60$ (C4):

$$M' = 2 \times 60 - \{60, 62, 64, 65, 67\} = \{60, 58, 56, 55, 53\}.$$

Each of the three sections has 16 bars; the conservation check is: $16 \times 3840 \times 3 = 184320$ ✓.

Worked example two: 100 bars. The total length is $100 \times 3840 = 384000$ ticks, and the sections are divided as $A(30)$–$B(30)$–$A'(30)$–Coda(10). Section A' is a recapitulation of section A an octave higher, and the Coda uses a combination of retrograde and inversion. Conservation check:

$$115200 + 115200 + 115200 + 38400 = 384000 \checkmark.$$

The generated result contains 216 notes in total, of which the three sections A, B and A' have 65 each and the Coda section 21. Over all 216 notes the interval sequence remains C4 → D4 → E4 → F4 → G4 unchanged, and the variation occurs only in the duration dimension; the ratios are given in Table 24.

Table 24: Ratio verification of the duration transformations (relative to a quarter note)

Group	Duration type	Duration (ticks)	Ratio
Group 1	quarter note	960	1 : 1 (baseline)
Group 2	half note	1920	2 : 1 (augmented 2 times)
Group 3	dotted quarter note	1440	1.5 : 1 (augmented 1.5 times)
Group 4	whole note	3840	4 : 1 (augmented 4 times)
Group 10	eighth note	480	0.5 : 1 (diminished 0.5 times)
Group 11	sixteenth note	240	0.25 : 1 (diminished 0.25 times)

Remark 6.8 (The limits of the verification). The two worked examples verify **the consistency of conservation and transformation**, that is, that the generated result satisfies the formal constraints of equations (17)–(18). They **do not constitute a verification of musicality**: this paper carries out no listening evaluation and no comparison with human compositional results. An isomorphic skeleton can carry mediocre or excellent music, and the skeleton itself does not distinguish between the two.

7 Corollary four: the coordinatization of semantics

This section sets out the fourth corollary: the relation between the human-readable position representation and the machine-executable absolute coordinate can be completely characterised by a pair of mutually inverse mappings. The so-called problem of "semantic translation" therefore closes here — it requires no understanding, only a bidirectional function.

7.1 Dual-coordinate-system comparison

Between the human perspective and the language-model perspective there exist two parallel coordinate systems: the former describes event blocks as "bar.beat", the latter performs its computation in absolute ticks. The two are related through two connecting lines — one bidirectional arrow indicating that the position representation and the absolute tick can be translated into each other in both directions through a protocol, with a constant and determinate mapping relation; and one mirror line indicating that the same musical event has a consistent spatiotemporal structure under both perspectives.

Below the two coordinate systems each has a route indicator line, representing respectively the editing route of a human using a DAW and the internal computation route of the language model; the two lines remain parallel, and always mark a common alignment position, indicating that the two stay time-synchronised at every instant.

Remark 7.1 (One observation awaiting verification). In the coordinate system of the language-model perspective, the trajectory of its computational inference shows a similarity in shape to the CC (continuous controller) curves that a human draws in a DAW for emotional expression. That similarity suggests that the two may be two different presentations of one and the same intrinsic musical structure. This observation is at present supported only by graphical intuition and has no quantitative verification, so it is recorded as a remark only.

7.2 The bidirectional mapping engine

Definition 7.2 (tick_translate)**.** The bidirectional mapping engine `tick_translate` takes a position representation and project parameters as input and outputs an absolute coordinate, or it takes an absolute coordinate and project parameters as input and outputs a position representation:

$$\texttt{tick_translate}(\text{Position}, \text{PPQ}, \text{TimeSig}) \to \text{AbsTick}, \tag{21}$$

$$\texttt{tick_translate}(\text{AbsTick}, \text{PPQ}, \text{TimeSig}) \to \text{Position}. \tag{22}$$

The engine has a built-in adaptation mechanism for time signature and resolution, so as to keep the coordinates consistent in any project context.

7.3 From semantic instructions to executable instructions

The inference results are encapsulated as structured semantic instructions for a language model to parse and execute. Taking "place MIDI number 60 (C4) at bar 2, beat 3, with a duration of 480 ticks" as an example, the mapping yields the executable form:

> at absolute tick coordinate 5760, place an event with `pitch=60`, `duration=480`.

This encapsulation layer is the key interface of the system described in this paper in the scenarios of AI music teaching, automated arrangement and intelligent composition.

Because this structured data already contains the complete position information of an event in the coordinate system together with its pitch semantics, a language model can understand an event's temporal position, duration, pitch and their mutual relations without any additional music-specific training, thereby lowering the training cost and the corpus threshold for music AI models.

8 Application: coordinate feedback in human–AI collaboration

The corollaries of the four preceding sections each hold on their own, but their value becomes fully apparent only once they converge. This section puts those corollaries back into a real human–AI collaboration scenario and examines one thing: when position, articulation and structure can all be measured by coordinates, **what happens to the language of feedback**. The teaching methods here are not an appendage but the operational form of this feedback mechanism. This section records one complete human–AI collaborative process, in order to show how coordinatized measurement works as a **language of feedback**. The case is a single-case observation and does not constitute a controlled experiment (see the explanation in section 3.9.4).

8.1 Setup

A comparison template was set up in a default project environment. In the first round no correction at all was made: the generative model was simply asked to write 16 bars of its own (4/4, starting from bar 1, beat 1); afterwards the 16 bars written by the author were supplied as a reference, and the model was asked to **compare them itself** rather than having the author point out where the errors were.

8.2 The difference report the model produced by itself

With no external hint, the model produced a structured comparison along six dimensions.

(i) Two-dimensional lattice distribution. The reference sample is dense and sparse by turns: bars 5–6 contain dense eighth notes and dotted rhythms, bar 15 is filled entirely by one whole note; the whole has an arch shape (the first half climbs from C4 to C5, the second half climbs from C5 to E6 and then falls back to B5); and it contains many non-standard durations, such as 1.25, 1.47, 1.85, 2.50 and 2.75 beats, whose landing points do not lie on the standard grid lines. The model described its own output as "almost nothing but quarter notes and half notes, evenly spaced", "the two halves are completely symmetrical, like an exercise", "not a single dotted note".

(ii) Diversity of durations. See Table 25.

Table 25: Comparison of duration diversity

Duration type	Reference sample	Model output
quarter note	present	many
dotted quarter note	occurs several times	none
eighth note	several rapid transitions	none
non-standard durations	1.25/1.47/1.85/2.50/2.75 beats	all standard durations
closing whole note	bar 15	none

The model attributed the difference as follows: non-standard durations can be converted exactly in mathematics (1.25 beats = 1200 ticks), and musically they are precisely the source of the "sense of breathing". **(iii) Phrase structure.** The model described its own output as having "only barlines — no phrase thinking that goes beyond the bar at all", whereas the motif of the

reference sample runs from `1.1.00` to `1.4.25`, spanning the whole of bar 1. In the model's words: this is not "the melody of bar 1" but "a complete musical sentence"; **the boundary of a melody is not the barline but the point where the vertical value changes**.

(iv) Pitch layout. The range of the reference sample is C4–E6 (two octaves + a major third), dense in the high register, with the chromatic note G#5 appearing in bar 14; the range of the model output is C4–C6, entirely diatonic.

(v) Summary of the core gaps. See Table 26.

Table 26: The core gaps as summarised by the model itself

Dimension	Model output	Reference sample
Duration diversity	uniform (quarter + half)	rich (dotted, eighth, non-standard durations)
Phrase structure	divided by barlines	cross-bar motifs + phrase hierarchy
Pitch layout	symmetrical and regular	arch shape + high-register breakthrough + chromatic notes
Rhythmic subdivision	none	several dotted and eighth-note transitions
Mathematical precision	standard durations only	every non-standard duration can be converted exactly into ticks

(vi) The model's conclusion. The original text is as follows.

> Through this melody you have perfectly demonstrated what "the fusion of mathematical logic and musical logic" means — behind every dotted note, every non-standard duration and every cross-bar phrase lies an exact tick computation, yet what emerges is not cold numbers but music with life in it. My melody is only a "correct sequence of notes"; your melody is music.

8.3 Observations

This case supports two observations, but both must be treated with caution.

Observation 8.1. ***Granularity of feedback.*** *When coordinatized measurement is available, feedback can be precise enough to say "the durations here are too uniform" or "this line lacks a cross-bar phrase", rather than "it doesn't feel right". The language of feedback shifts from aesthetic judgement to a localisable structural description.*

Observation 8.2. ***The nature of the model's behavior.*** *The model was not told the direction of improvement directly; it extracted its own dimensions of comparison, such as slope, amount of breathing and register control. This suggests that what it did was to* ***reconstruct*** *an answer on top of existing logic rather than to invoke an existing template.*

Remark 8.3 (The limits of the strength of the evidence). This section is a single-case observation, with no control group and no independent raters. The model's self-report may be influenced by the wording of the prompt, and its judgements of "correct" have not been verified externally.

The conclusions of this section should therefore be treated as **case evidence**, used to show the usability of the method in practice, and are insufficient to support a universal claim about its effects.

8.4 The teaching pipeline: a nine-layer framework

The collaborative process of section 8 can be summarised as a nine-layer pipeline, see Table 27. Its design starting point is the same as that of this paper: to translate musical logic into a computable intermediate layer, so that teaching instructions need not depend on vague terminology.

Table 27: The nine-layer structure of the teaching pipeline

Layer	Name	Action on the teacher side	Action on the model side
1	general language layer	describe a feeling in everyday language	receive a vague instruction
2	translation layer	turn the feeling into concrete rules	establish a "concept → rule" mapping
3	structuring layer	organize scattered rules into a system	establish relations among the rules
4	guided-learning layer	ask, guide and prompt rather than give the answer directly	discover the problem on its own under guidance
5	mathematisation layer	turn rules into coordinate parameters	the rules become executable instructions
6	demonstration layer	provide a "right" sample	compare its own output with the sample
7	error-correction layer	point out "this is wrong" without saying "how to change it"	analyze the gap itself and correct it on its own initiative
8	iteration layer	repeat the correction process	approach the correct result step by step
9	confirmation layer	confirm whether the understanding is correct	lock in the standard of "that is right"

Layers 4 and 7 are the key difference between this framework and conventional supervised fine-tuning: both require the feedback provider **not to give the answer directly** but only to give a coordinatized description sufficient for the trainee to locate the gap by itself.

8.5 Comparison of seven teaching methods

In practice the pipeline above borrows the structure of seven existing teaching methods; the correspondence is given in Table 28.

The withdrawal path of scaffolding deserves a separate record, because it gives an operable sequence:

Table 28: The teaching methods actually used in the collaborative process

Teaching method	**Intervention mode**	**Corresponding action in this case**
Socratic method	guided questioning	"you compare them first, then I'll tell you"
scaffolding	gradually withdraw support	from "giving the structure" to "giving the direction" and then to "giving an adjective"
cognitive apprenticeship	demonstration → imitation → working together → independence	"I write one, you write one, and we compare"
problem-based learning	driven by a problem, the knowledge emerges	"how do you localize exactly in 6/8"
learning by doing	form understanding within a real task	"just write 16 bars; when you're done I'll tell you what's wrong"
generative learning theory	let the trainee construct it itself	"write another one according to these three rules"
guided-text method	give the problem and the steps	"work on the Y axis"

"I give the structure" → "I give the rules" → "I give the direction" → "I give one adjective". When the trainee can generate 16 bars from the adjective "warm and bright" alone, the scaffolding has already been withdrawn.

Remark 8.4. All the teaching methods listed in this section are existing theories in pedagogy; this paper proposes no new teaching method, it merely records how they were combined in one particular collaborative process. Their generalisability still requires verification through repeated experiments.

8.5.1 Potential implications for music education systems

The core chain of traditional music training is "score symbols → auditory experience → performance expression", whose priority is that experience and feeling come before logic. Once position and duration are coordinatized, a parallel structural layer appears alongside that chain; the two are compared in Table 29.

It must be stressed that this is not a relation of replacement. What traditional teaching trains is perception and expression, whereas what coordinatization supplements is a computable underlying structure: things that could previously be understood only by feeling can now be described precisely. The two address problems at different levels.

8.5.2 Potential implications for the assessment system

If position can be measured, then several items in music assessment that previously depended entirely on subjective judgement can acquire an objective dimension of reference, see Table 30.

Table 29: Comparison of the traditional route and the coordinatized route

Traditional route	**Coordinatized route**
the time signature is “defined”	the time signature is “derivable”
durations are “heard”	durations are “computed”
position is “aligned by feel”	position is an “exact coordinate mapping”
rhythm is “groove training”	rhythm is the “temporal distribution of the event chain”

Table 30: Assessment items that can be made objective

Assessment item	**Traditional manner**	**Coordinatized manner**
whether note positions are accurate	judging that it is “about right”	checking whether the absolute tick falls at the expected position
whether rest durations are correct	listening to judge whether the silence is “right”	measuring whether the tick length of the rest equals the expectation
whether the rhythm is stable	judging from experience	computing whether the tick intervals between adjacent notes are even
whether the melody is complete	relying on a “sense of phrasing”	checking whether the event chain is continuous and free of unexpected interruptions

Remark 8.5. Table 30 gives a **reference**, not a **verdict**. The coordinate system itself does not answer “does it sound good”; it answers only “precise or vague”, “consistent or random”, “structurally complete or broken”. Measuring the measurable part is not the same as claiming that the unmeasurable part has no value.

9 Implementation and validation

9.1 System composition

Following the definitions of sections 3–7, the system consists of four parts: the absolute spatiotemporal coordinate system, the TPB preset mechanism, the bidirectional mapping engine, and the semantic encapsulation layer.

9.2 Misalignment of the counting origin

One systematic deviation occurred during measurement: with the anchor given as `2.1.00`, the coordinate of the first event nevertheless landed on `3.1.00`, a deviation of exactly one bar. The cause was that the absolute tick of the left edge of the material had already been treated as the aligned baseline and was then added a second time. The fix is to add the counting-origin offset back explicitly according to equations (5)–(6), and to apply that offset only once in the whole pipeline.

9.3 Scope failure

The second class of defect is more insidious: when an outer variable is modified through a closure, if it is not declared explicitly, the assignment creates a new local variable of the same name inside the function scope and the outer variable remains undefined. The fix is to let the function only compute and return a value, with the outer level performing the assignment.

```
def get_anchor_wrong():
    SM = compute()      # creates a local variable; the outer SM is still undefined
    return None

def get_anchor():
    return compute()    # only computes and returns

SM = get_anchor()       # received by the outer level; the scope is clear
```

9.4 Experimental setup and results

The experiments were carried out in the DAW with a pulse base of PPQ = 960; the time signatures were set one by one and the read-outs were compared with the script output. The results are given in Table 31.

Table 31: Parameters and consistency under different time signatures

Time signature	BPI	Denom	TPB	Position	Absolute tick	Consistent
4/4	4	4	960	`2.1.00`	3840	✓
4/4	4	4	960	`3.1.00`	7680	✓
6/8	6	8	480	`4.1.00`	8640	✓
6/8	6	8	480	`5.3.00`	12480	✓
7/4	7	4	960	obtained automatically	—	✓

The last row of Table 31 deserves a separate remark: 7/4 was not set in advance; the program read BPI = 7 and Denom = 4 automatically from the project metadata, and the localization result is still correct. This shows that equation (2) is a general form containing no preset branch for any particular time signature.

10 Discussion

10.1 The nature and the position of this work

It is necessary to state the nature of this work truthfully. The mathematical essence of the core coordinate system is the **piecewise evaluation of a piecewise linear function**, which is trivial content; the only technical point is the relativisation of the bar index within a segment $(M - A_s)$, and there is no non-trivial proof. It produces no new mathematical knowledge; rather, it fixes a known but unformalised engineering process into an invocable specification.

It is nevertheless still worth proposing separately, because that specification **did not previously exist in explicit form**: DAW software implements it internally but exposes only "bar.beat";

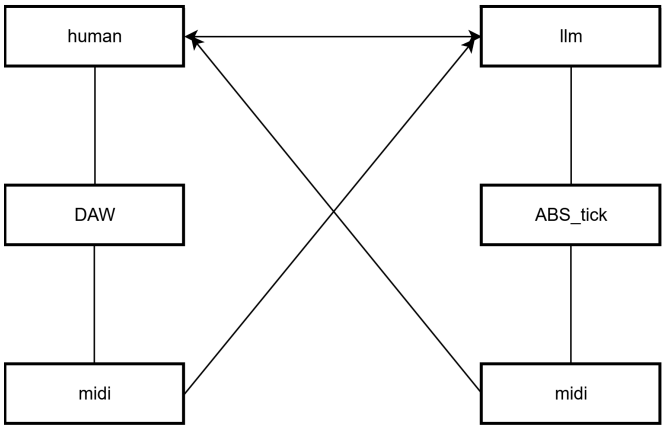


Figure 9: The coordinate round trip in the human–AI interaction path.

writing that mapping as a closed form, giving its degenerate property and explicitly drawing its applicability boundary is the work done in this paper. The accurate statement is therefore: this coordinate system is not a mathematical discovery but an $O(1)$ computational interface with clearly defined boundaries, whose upper bound of value is determined by the time-signature change density (section 3.9).

10.2 Relation to existing representation systems

The coordinate system of this paper does not replace the MIDI message stream or notation description languages; it supplies the mapping layer between the two. MIDI takes performance events as its unit of record, MusicXML takes the layout of the score as its object of description, and neither provides an absolute position quantity independent of the time signature; the absolute tick defined in this paper is exactly that missing quantity.

10.3 Potential impact on the formulation of music theory

This paper does not claim that numbers should replace description, but it points out one actual change: several concepts that could long be expressed only qualitatively acquire, after coordinatization, a form that can be **counted pair by pair** — the amount of overlap, the pitch difference and the amount of gap (section 5.1). In existing accounts these quantities rely on overall impressions such as “a bit more legato here” or “a bit more space here”, and overall impressions are hard to compare and hard to reproduce. Once they have a quantitative form, they can at least be compared, reproduced and modified locally.

10.4 Limitations

The scale of the experiments in this paper is limited, and the conclusions are constrained by the following conditions.

1. Verification was completed on a single DAW only, and no cross-validation against other implementations has been carried out.
2. No strict proof of the upper bound of the rounding error under a changing time signature has been given; only experimental observation is available (section 3.7).

3. The correctness of the "equidistance exclusion method" in voice separation when three or more voices cross simultaneously has not been proved.
4. The effectiveness of the semantic encapsulation layer in real teaching scenarios has not been verified by user studies.
5. The collaborative case in section 8 is a single-case observation with no control group and no independent raters, and its conclusions should be treated as case evidence rather than a universal claim.
6. The two worked examples of the generation architecture in section 6 verify only the consistency of the formal constraints; no listening evaluation was carried out, and they constitute no claim about musicality.

10.5 Future work

The following problems have a clear direction but were not resolved in this paper; they are listed in order of priority.

1. Give an upper bound for the rounding error under a changing time signature, or prove that the error is identically zero for particular values of PPQ.
2. Generalise the equidistance exclusion method to the case of several voices crossing simultaneously, or characterise its failure condition.
3. Reproduce all the experiments on a second DAW, in order to test the implementation-independence of the conclusions.
4. Design a listening experiment with a control group, in order to assess the actual musical effect of the proposed generation architecture.

11 Conclusion

Starting from an observation of the DAW interface, this paper constructs the absolute spatiotemporal coordinate system MATCS. The main conclusions are:

1. The position of a musical event can be translated into an absolute tick value independent of the time signature, the tempo and the tonality, whose form is given by equation (2);
2. "Irregular meter" and "variable time signature" have different properties in tick mathematics, and the latter is the real cause of the failure of the original formula; under a changing time signature the original formula must be generalised to the piecewise summation form (10), of which the original formula is the degenerate special case at $n = 1$;
3. Taking "the instant at which the vertical value changes" as the sole criterion and redefining rests as empty events unifies the localization logic of notes and rests, and no undefined interval remains on the time axis;
4. The applicability domain of that form is constrained by the time-signature change density: the parameter count grows linearly, and as the density approaches one change per bar the method must switch to a time-signature event table plus binary search;

5. The reverse mapping cannot be replaced by hand calculation, and this is where the form is genuinely irreplaceable.

Truthfully, the mathematical content of this paper is not substantial: its essence is the piecewise evaluation of a piecewise linear function, the only technical point is the relativisation of the bar index within a segment $(M - A_s)$, and there is no non-trivial proof. Its contribution lies in fixing a known but unformalised engineering process into a specification, and in clearly drawing its applicability boundary.

A Excerpts from the research log

This appendix excerpts, in chronological order, the original records made during the derivation; several of the criteria in the main text arose from them.

> **Day one · stability of the scale**. Observe the "bar.beat.tick" read-out in the DAW over a long period: whether the time signature is modified, the tempo changed or the material dragged, that read-out never changes and always points to the same absolute position.
>
> **Day two · an attempt at formalisation**. The rudimentary form of equation (2) was obtained, and the first beats of the second and third bars under 4/4 were substituted for checking, giving 3840 and 7680, in agreement with the software read-out.
>
> **Day three · the counterexample**. After the time signature was changed from 4/4 to 6/8 the result showed a significant deviation, and dragging the material or moving notes a long way failed in the same manner. Inspecting the deviation data revealed a **determinate** error value between the two time signatures — the error is determinate rather than random, which shows that the model is not wrong but is missing a variable.
>
> **Day three · the key observation**. An eighth note under 4/4 has a span of `0.0.50` in the read-out, while the same eighth note under 6/8 reads `0.1.00`. The two have the same tick length (480) but are interpreted as different numbers of beats. The conclusion follows: the duration scale is not a global constant but a local variable determined by the denominator of the time signature.
>
> **Day four · global localization**. Treat the DAW grid as a plane Cartesian coordinate system: the horizontal axis is already given by the absolute tick, while the vertical axis is still to be established. The difficulty of the vertical axis lies not in labelling but in the criterion — how to decide where a note begins and ends. This yielded "event = the instant at which the vertical value changes".
>
> **Day five · self-positioning**. Looking back at the form itself: its mathematical essence is the piecewise evaluation of a piecewise linear function, which is trivial content; the only technical point is the relativisation of the bar index within a segment; and there is no non-trivial proof. Its value lies not in producing new knowledge but in fixing a known process into an invocable specification.

B Implementation of the core algorithm

Construction of the piecewise axis and the bidirectional mapping (the original implementation was in Lua; here it is transcribed into an equivalent form):

```python
def build_axis(sections, ppq):
    """sections: [(upToBar, bpi, denom), ...]; upToBar is strictly increasing"""
    axis, acc_bar, acc_tick = [], 0, 0
    for up_to, bpi, denom in sections:
        bars = up_to - acc_bar
        tpb = round(ppq * 4 / denom)
        axis.append(dict(from_bar=acc_bar + 1, to_bar=up_to, bars=bars,
                         bpi=bpi, tpb=tpb, start_tick=acc_tick,
                         ticks_per_bar=bpi * tpb))
        acc_bar = up_to
        acc_tick += bars * bpi * tpb
    return axis

def pos_to_tick(axis, bar, beat):
    for seg in axis:
        if bar <= seg["to_bar"]:
            rel = bar - seg["from_bar"]
            return seg["start_tick"] + (rel * seg["bpi"] + (beat - 1)) * seg["tpb"]
    raise ValueError("bar out of axis range")

def tick_to_pos(axis, tick):
    for i, seg in enumerate(axis):
        seg_end = seg["start_tick"] + seg["bars"] * seg["ticks_per_bar"]
        if tick < seg_end or i == len(axis) - 1:
            local = tick - seg["start_tick"]
            bar_off = local // seg["ticks_per_bar"]
            rem = local - bar_off * seg["ticks_per_bar"]
            beat = rem // seg["tpb"] + 1
            return seg["from_bar"] + bar_off, beat
    raise ValueError("tick out of axis range")
```

C Glossary of terms

Table 32: Core terminology

Term	Meaning
MATCS	Absolute Spatiotemporal Coordinate System, the position representation proposed in this paper
absolute tick	the tick count from the start of the project to that event, independent of the time signature, the tempo and the tonality
empty event	an event whose pitch attribute is empty and which has explicit onset and offset times; a rest is one example
event chain	the uninterrupted sequence formed by the end of each event meeting the start of the next

(*continued*)

Term	**Meaning**
TPB	the number of numerical coordinate units per beat, = PPQ × 4/Denom
BPI	Beats Per Bar, the number of beats per bar, equal to the numerator of the time signature
PPQ	Pulses Per Quarter Note, the system's preset time resolution
Denom	the denominator of the time signature, which determines what kind of note "one beat" corresponds to
simple meter	2/4, 3/4 etc., a single group of accents
compound meter	6/8, 9/8, 12/8, each "beat" being a dotted note
irregular meter	5/8, 7/8, unequal division within the bar, the time signature itself unchanged
variable time signature	the signature is changed mid-piece and must be handled segment by segment
segment table	the prior metadata recording n_k, p_k, d_k for each time-signature segment
segment start S_s	the cumulative tick boundary at the start of segment s
anchor	the reference position chosen when measuring relative distances
vertical value	the pitch state at a given absolute tick, 0 denoting a rest
tick_translate	the bidirectional mapping engine between the position representation and the absolute coordinate
line_for_llm	the structured semantic encapsulation layer for language models